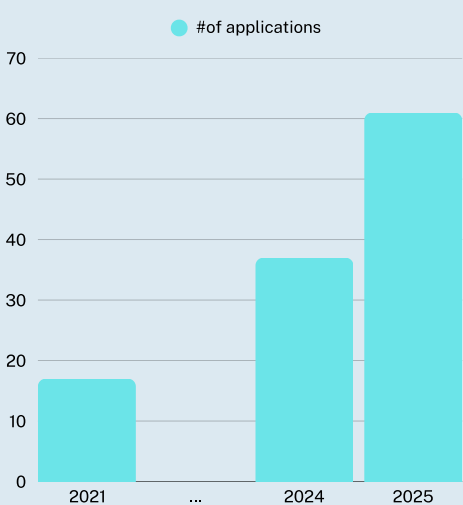


1 About UCSD Sea++

Team Sea++ brings students across multiple disciplines and years together, cultivating an interest in high performance computing at the undergraduate level. The team is under UCSD's Supercomputing Club, which hosts various projects and workshops for the undergraduate engineering community at UCSD as well as the annual Single Board Cluster Competition.



Diverse team and applicant pool:

- Exponential rise in applications
 - ~2x last year and ~4x 2021
- Team is an **interdisciplinary mix** of students studying computer science, computer engineering, electrical engineering, bioinformatics, and mathematics.
- 3/6** students identify as female and are supported by SDSC's Women in HPC chapter



Gauri Renjith
Captain / MLPerf

Gauri Renjith

3rd year Computer Science/Bioinformatics
Role: Captain, MLPerf Lead
Gauri is passionate about the intersection of machine learning, HPC and biological data. She optimized molecular dynamics software at SCC24 and is excited to return this year!



Ferrari Guan
Sysadmin / HPL-MxP

Ferrari Guan

3rd Year Computer Engineering
Role: Sysadmin, HPL-MxP Lead
Ferrari's interests lie in microcontrollers/embedded systems, Mechanical Computer-Aided Design, Printed Circuit Board (PCB) design, computer security, computer infrastructure, and operating systems. He was an alternate for the UCSD SCC24 team, where he worked on systems and networking.



Luiz Gurrola
Sysadmin / HPL

Luiz Gurrola

4th Year Math-Computer Science
Role: Sysadmin, HPL Lead
Luiz is interested in all aspects of Computer Science with special interests in Digital Design, HPC, and Operating Systems. Luiz is constantly looking to improve his homelab and system administrator skills.



Cecilia Li
Reproducibility

Jinru (Cecilia) Li

2nd Year Electrical Engineering
Role: Reproducibility Lead
Cecilia is excited to explore both the hardware and software components of HPC. She has hands on experience with hardware architecture research, sysadmin, and networking.



Shing Hung
SST

Shing Hung

4th Year Computer Engineering
Role: SST Lead
Shing has diverse experience across embedded systems, hardware design, and computer science, specifically in PCB designs, microcontroller integration, and full stack website development.

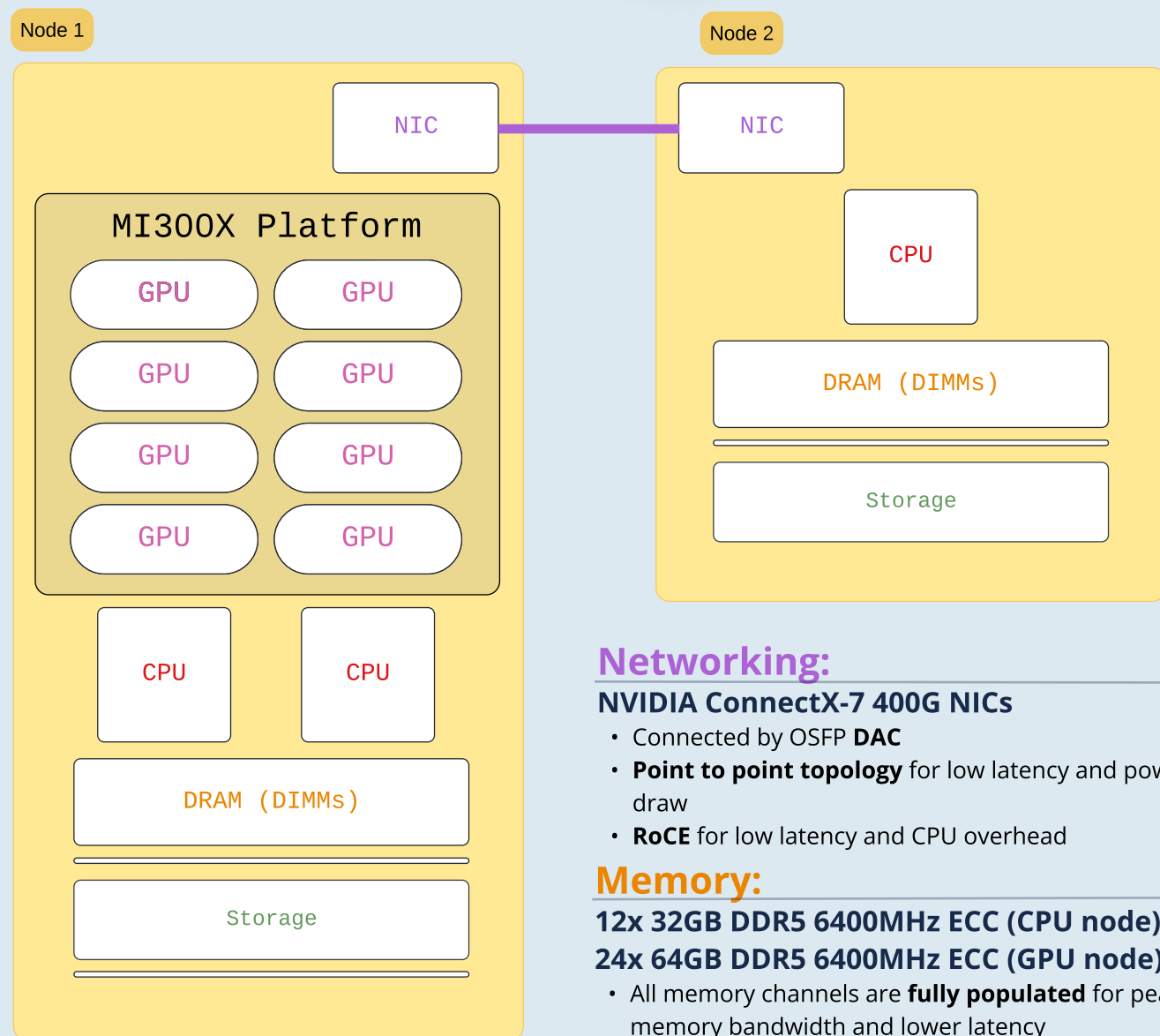


Ryan Estanislao
ECE

Ryan Estanislao

2nd Year Computer Science
Role: ECE Lead
Ryan has experience in machine learning and cybersecurity, and is further interested in the combination of computer science, data analytics, and econometrics.

2 Hardware Setup



CPU:

AMD EPYC Generation 5 Turin (9755 & 9745, 128C)

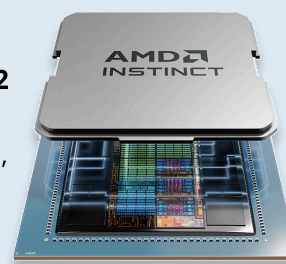
- 12 memory channels with **6400 MT/s, 614 GB/s** Bandwidth
- Zen 5** architecture = higher frequencies and faster DRAM
- Node 1 has **2 AMD EPYC 9755s**, Node 2 has **1 AMD EPYC 9745**



GPU:

8x AMD MI300X OAM GPUs

- Quick memory transfer - **5.3 TB/s**
- Easily handles large amounts of data - **192 GB HBM RAM** per GPU
- Strong performance - **2614.9 FP8 TFLOPS**, 1307.4 FP16 TFLOPS, 304 GPU compute units



AMD MI300X Platform

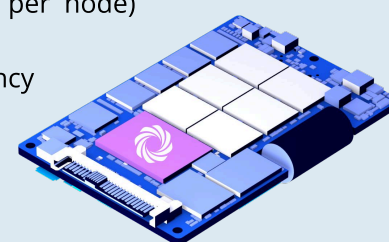
- Baseboard provides **128 GB/s** links between any 2 GPUs

Storage:

ScaleFlux CSD 3310 7.68TB (2 per node)

- Hardware data compression
- 300x higher compression efficiency and 2x better storage IO

Samsung PM9A3 1.9TB M.2 NVMe (2 in GPU node)



3 Software Stack

OS: Rocky Linux

- Free and open source**
- Familiarity** from mentors and team members via projects

Filesystem: NFS

- Minimal setup, reasonable for small scale cluster
- Access to files from either node

Deployment & Management: Ansible Scripts

- For infrastructure setup and dependency installation
- Ensures **system and deployment reproducibility**
- Simplifies system setup and resets

Monitoring: Grafana

- Prometheus backend for metric storage

Tools:

GNU Tools, HIPCC, ROCm, Docker, Anaconda, OpenMPI



4 Application/Benchmark Optimization

HPL and HPL MxP

Building HPL/MxP:

- Adapt the build script from **AMD Infinity Hub** for compatibility with our system's architecture and configuration

Running HPL:

- To achieve optimal performance, we follow the **recommended parameter configurations** from AMD Infinity Hub.
- For multi-node running, we need to test all the settings and find the **best asymmetric parameters** for our hardware setup.

Running HPL-MxP:

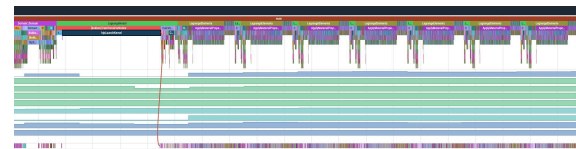
- To achieve the optimal performance, we refer to the parameters and sizes provided in the AMD Infinity Hub.
- To get the best multi-GPU run, we will rigorously test all the parameters options for our specific hardware architecture.



MLPerf

- AMD provides **containerized setup** for MI300X nodes
 - utilizes **FP8 quantization**
 - explore **supported backends** (Ray, SGLang) and attention mechanisms if *time permits*
- We will:

- test adjustable **performance environment variables**
- test increased % of **GPU memory utilization**
- profile GPU usage** and efficiency with **ROCm Profile Data tool**



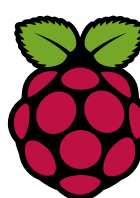
Time and Power Strategies

Time

- Coordinate applications and benchmarks runs** between teammates
- Schedule **mystery app runs** for the end of the 48 hour period

Power

- Raspberry Pi** to monitor and log power and data
- Grafana dashboard** to monitor power draw of each node, **webhooks** to send alerts
- Manually lower the max TDP** of the GPUs if going over power



SST

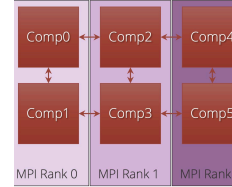
- Build and containerize SST with **Docker** for portability

AMD CPU-specific strategies

- AOCL library** and architecture-specific flags to leverage hardware
- Memory/NUMA optimizations**

OpenMPI runtime process tuning:

- Binding ranks** to specific cores and sockets
- Testing OpenMPI flags to improve communication
- Improve **cache locality**



Climate Emulation

- Install **HIP-enabled dependencies** (ex. StarPU, Chameleon) and ROCm math libraries to accelerate the linear algebra that underlies the emulator

- Profile with **ROCm tools** and adjust StarPU and Chameleon parameters on AMD's HPC Fund

- Explore and understand covariance approximations and how to take advantage to lower compute costs



Reproducibility Challenge

- Aim to understand the **two kernels**
- Compare **FA-BSP** implementation vs **BSP (Ripples)** implementation
- Build **IMM HClb** package and reproduce **figures** prior to competition for familiarity
- Run the provided graphs to collect:
 - Runtime** metrics
 - Scalability** metrics
 - Memory** metrics
 - Quality** metrics
- Document hardware and software stack** extensively to streamline paper writing

Our Competitive Advantage

Our team is extensively supported by the University of California, San Diego (the Department of Computer Science and Engineering in particular) and the San Diego Supercomputing Center. With access to SDSC's Expanse system and vendor research clusters to prepare and supportive faculty mentors and competition alumni, **we are confident that we can bring out the best of our team and our cluster.**

